

SECURE SCAN DESIGN WITH ISOMORPHIC REGISTERS

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Abstract: In this paper, we first introduce Isomorphic Redundancy concept. Two functionally equivalent shift registers can be Isomorphic to each other. They can be equivalent to each other by simple permutation of states in state tables. Two Isomorphically redundant circuits can be used to prevent two bit change insertion attack. In, addition we also propose a new model with the help of functionally equivalent shift registers. It is highly non-linear and scan-secure model.

Keywords: Isomorphism, Equivalence, LFSR, LF²SR, 8X1 MUX, Testability, Shift registers, Functionally equivalent, Extended de Bruijn graphs.

INTRODUCTION

Testability is a very important property of every hardware device allowing its user to check if the device operates correctly. Design for Testability (DFT) is a technique that is used to test digital circuits for faults. Faults in digital circuits are against the security of a chip. Increasing Complexity in VLSI design is a remarkable concern on the reliability of digital circuits. Scan registers or scan chains are proven to be effective in improving the testability of digital circuits. But this scan chains can be used to steal important information such as intellectual property (IP) and secret key of cryptographic chips. Scan chain based side channel attack is also possible. So there's a need for an efficient solution so that it can satisfy testability and security both. Several ideas have been proposed which aims to satisfy both testability and security. A scan-chain design based on scrambling was proposed in [3], where flip-flops are dynamically recorded in a scan chain. Another concept is given in [4], where secure scan-chain architecture with mirror key register (MKR) was introduced. This architecture has both test/normal mode (insecure) and normal mode (secure). Switching back from secure mode to insecure mode is only done through a power off reset. This register works like normal registers during insecure mode and test vectors can be scanned in and the result can be scanned out.

When the circuit is in secure mode, this registers load the secret key information and the contents of mirror key registers can't be scanned out until being reset. Another similar scheme which has also insecure and secure modes is the lock and key security technique proposed in [5]. But this technique divides the scan chain into smaller subchains of equal length and uses test security controller (TSC) to switch between secure and insecure modes. Vlm-scan that utilizes some flip-flops in a scan chain for authentication to move to test mode [6]. It claims to provide superior method compared to ones mentioned. If the proper sequences of test keys are scanned in to the used flip-flops then only the circuit can proceed to test mode. Another advantage is the test controller can be tested. But a long test key sequence is still needed. There is another model in [7] called secured flipped scan-chain, which uses inverters in the scan chain for protection. With additional NOT gates this architecture is tested the same way with scan chains. Having sufficient number of inverters, their positions can still be determined by simply scanning out after resetting (to zero) all the flip-flops in the scan chain. Thus, the internal state can be identified. All of these proposed techniques except last one add extra hardware outside of the scan chain. This entails several disadvantages such as high area overhead, timing overhead or performance degradation, increased complexity in testing and limited security for the registers part among others. But the scan design approach by using extended shift registers that are functionally equivalent but not structurally equivalent to shift registers requires little area overhead and no performance overhead for normal operation [1], [2]. In this paper we have introduced Isomorphic redundancy concept using functionally equivalent LF²SRs. And also proposed an Isomorphic circuit using two functionally equivalent LF²SRs. It can be used to prevent two bit change insertion attack. Also introduced a new model using functionally equivalent shift registers. And this model is scan secure.

ISOMORPHIC REDUNDANCY

We have considered two functionally equivalent LF²SRs R_1 and R_2 , as shown in Fig. 1 and Fig. 3 respectively. We can say they are Isomorphic to each other, i.e. by simple permutation of states in the state table. We can easily make them equivalent to each other, as shown in Fig. 5.

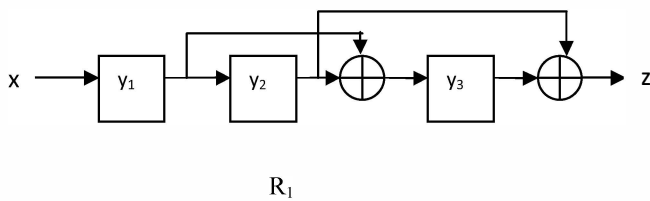


Fig. 1 Functionally equivalent LF²SR R_1

States	0	1
$S_0(000)$	$S_0/0$	$S_4/0$
$S_1(001)$	$S_0/1$	$S_4/1$
$S_2(010)$	$S_1/1$	$S_5/1$
$S_3(011)$	$S_1/0$	$S_5/0$
$S_4(100)$	$S_3/0$	$S_7/0$
$S_5(101)$	$S_3/1$	$S_7/1$
$S_6(110)$	$S_2/1$	$S_6/1$
$S_7(111)$	$S_2/0$	$S_6/0$

Table 1: State transition table for R_1

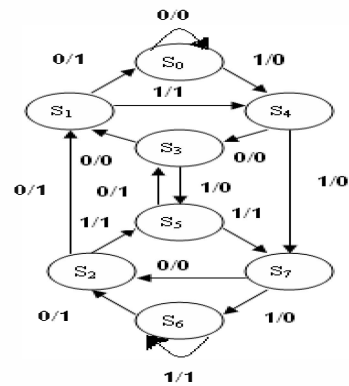


Fig. 2 Extended de Bruijn graphs of R_1

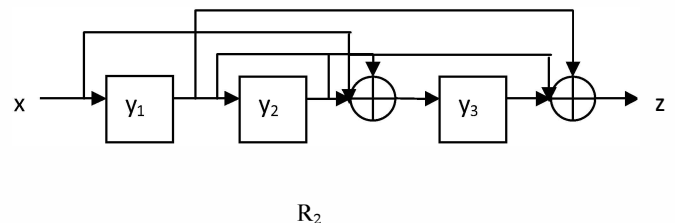
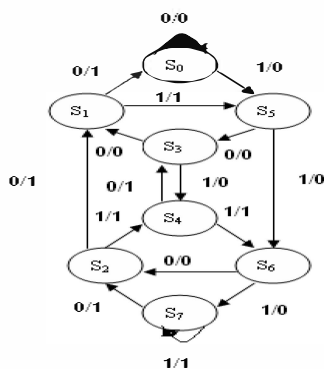
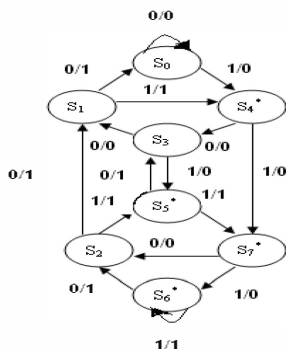


Fig. 3 Functionally equivalent LF²SR R_2

States	0	1
$S_0(000)$	$S_0/0$	$S_5/0$
$S_1(001)$	$S_0/1$	$S_5/1$
$S_2(010)$	$S_1/1$	$S_4/1$
$S_3(011)$	$S_1/0$	$S_4/0$
$S_4(100)$	$S_3/0$	$S_6/0$
$S_5(101)$	$S_3/1$	$S_6/1$
$S_6(110)$	$S_2/1$	$S_7/1$
$S_7(111)$	$S_2/0$	$S_7/0$

Table 2: State transition table for R_2

States	0	1
$S_0(000)$	$S_0/0$	$S_4^*/0$
$S_1(001)$	$S_0/1$	$S_4^*/1$
$S_2(010)$	$S_1/1$	$S_5^*/1$
$S_3(011)$	$S_1/0$	$S_5^*/0$
$S_4^*(101)$	$S_3/0$	$S_7^*/0$
$S_5^*(100)$	$S_3/1$	$S_7^*/1$
$S_6^*(111)$	$S_2/1$	$S_6^*/1$
$S_7^*(110)$	$S_2/0$	$S_6^*/0$

Table 3: Equivalent state table for R_2 **Fig. 4** Extended de Bruijn graph of R_2 **Fig. 5** Equivalent Extended de Bruijn graph of R_2

Thus we can make R_2 equivalent to R_1 if we interchange the states S_4 with S_5 and S_6 with S_7 , i.e. by renaming $S_5 \rightarrow S_4^*$, $S_4 \rightarrow S_5^*$, $S_7 \rightarrow S_6^*$ and $S_6 \rightarrow S_7^*$, as shown in Table 3 and Fig .5. Only R_1 and R_2 are not Isomorphic to each other, but we can take any two functionally equivalent LF^2SRs to make them Isomorphic to each other, by allowing simple permutation of states in state table.

ISOMORPHIC CIRCUIT DESIGN

Two functionally equivalent registers can also be Isomorphic to each other by a simple permutation of the states in the state table. We can easily make them equivalent to each other. If two Isomorphically redundant registers are used then we can prevent single bit change insertion attack on both of the extended shift registers. A bit change is implemented simultaneously at both the registers and hence the differential value is revealed at the output at the same time. Therefore, the provision is that if the two input bits are changed, differential values at the output cannot determine the cause due to which register the value has changed. In the Fig. 6, we have used two functionally equivalent LF^2SRs and output of each of the register is in a D-flip flop and outputs of two D flip-flops are in an OR gate. Their output sequence is identical to each other for the same inputs. With D-flip flops the outputs are indistinguishable at any instant. When the registers are interchanged, the state table and state diagram remains the same if there is a simple permutation of states are allowed i.e. S_1 and S_2 are interchanged, i.e. by simply renaming S_1 as S_2 . This happens when these two registers are respectively in 000/011, 001/010, 100/111 and 101/110 initial states. We have considered Table 4 and Fig. 7 when these two registers are in 000/011 initial state (0, 1, 2 and 3 are decimal equivalents of X_1 and X_2 and S_0, S_1, S_2 and S_3 are initial states of D_1D_2 with decimal equivalents of 0,1,2,3 respectively), then the state table and state diagram are remains unchanged if S_1 and S_2 are interchanged. This is true for any two functionally equivalent LF^2SRs .

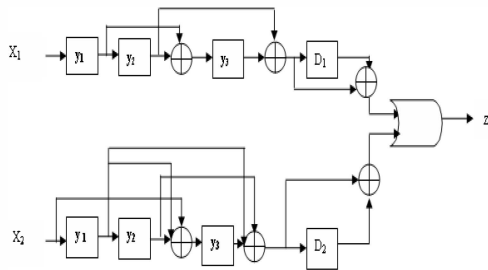


Fig. 6 Isomorphic circuit

	0	1	2	3
S_0	$S_0/0$	$S_0/0$	$S_0/0$	$S_0/0$
S_1	$S_0/1$	$S_0/1$	$S_0/1$	$S_0/1$
S_2	$S_0/1$	$S_0/1$	$S_0/1$	$S_0/1$
S_3	$S_0/1$	$S_0/1$	$S_0/1$	$S_0/1$

Table 4: State transition table when initial state either 000 or 011

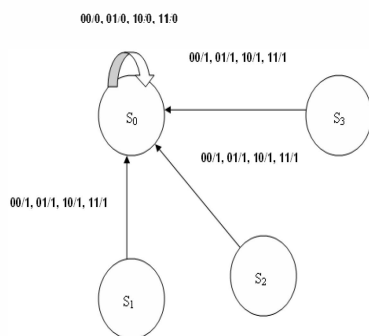


Fig. 7: State diagram when initial state either 000 or 011

PROPOSED MODEL

We have used 8 different functionally equivalent LF^2SR , LFSR and an 8X1 MUX in the proposed model shown in Fig. 8. Some dummy flip-flops (shaded flip-

flop in Fig. 8) were used here to prevent single bit change insertion attack and to make the design more scan secure. Outputs of each 8 extended shift registers were given to an 8X1 multiplexer. From input each single bit goes to 8 different extended shift registers and all the shift registers are indistinguishable [1]. A functionally equivalent shift register, whose output of each flip-flops were used as select line of the 8X1 MUX. We have considered that all the shift registers are in 000 initial state. For single bit change insertion attack value changes at the output will be same for all the functionally equivalent shift registers [1]. If we don't have idea about connection information then it is not possible to find out initial state of those extended shift registers from input/output sequences. In this described model each single bit is going through 8 different paths. So for a given input, output will be changed such way that one can't predict the initial states of extended shift registers. Therefore, we conclude that the model is scan-secure.

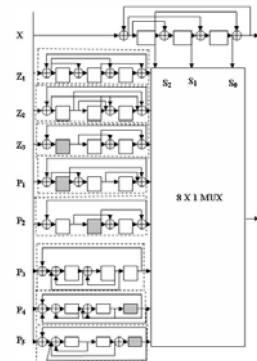


Fig. 8 Proposed model

EXPERIMENTAL RESULT OF THE PROPOSED MODEL

Input 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Output 0 0 0 1 1 1 1 0 0 0 0 1 0 0 1 0 0 0 1

No. Of toggles	=	7
Non linearity	=	7^2
Percentage non-linearity	=	35%.
No. Of changes	=	13
Instability	=	65%

From input to output there are 13 numbers of changes, so it is not possible to find out initial state of all the extended shift registers from input/output sequences. Where we know that the structure of extended scan register (ESR) is important to the attacker in order to understand the scanned-out values from the registers, which can reveal the key stream or the initial values of the internal registers, once computed. Thus the level of the security is dependent on the difficulty of determining the ESR structure. As the model is highly non-linear so this proposed model is scan-secure.

CONCLUSION

A new model has been introduced. It is a highly non linear scan secure model. From input to output there are so many changes that one cannot find out initial state of all extended shift registers if the connection information is secret. A new isomorphic circuit is introduced here using two functionally equivalent shift registers and two D flip-flops. This Isomorphically redundant circuit can be used to prevent two-bit change insertion attack.

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